

AI FOR GOOD - HACKATHON 4: PICTURE THE PLANET

ComfyUI Video Generation Technical Pipeline Documentation & Architecture Breakdown & prompts we used

1. Generative Video Pipeline (MiniMax H3 Model)

This workflow handles the generative synthesis of individual video shots for the climate action short film using the MiniMax H3 (Hailuo) neural architecture.

- **Nodes Used:** Resolution Selector (calculates and enforces a consistent 16:9 aspect ratio at 608 × 352 pixels, 0.2 megapixels, multiple 32 to ensure hardware stability on local consumer GPUs); Image to Video (MiniMax H3) as the core generation engine combining prompt instructions, motion parameters, and neural weights; Save Video for encoding and exporting the generated 4-second clip locally as an MP4 container.
- **Models & Weights:** Diffusion UNet (minimax_h3_fl2va_pruned_int8_convrot.safetensors, quantized for optimized local inference); Text Encoder (qwen3vl_32b_minimax_h3_nvfp4_awq.safetensors); Video VAE (minimax_h3_video_vae_int8_convrot.safetensors); Audio VAE (minimax_h3_audio_vae_fp32.safetensors); Accelerator LoRA (minimax_h3_fl2v_turbo_8step with 8 sampling steps and 0.80 model strength for physically realistic motion without geometric distortion).
- **Rubric & Assessment Relevance (K1 / Working Prototype):** Demonstrates that every visual shot is synthesized directly from an end-to-end, fully reproducible ComfyUI pipeline with fixed, verifiable seeds (e.g., 757358688076807).

The screenshot displays a ComfyUI workflow for video generation. The main node is 'Image to Video (MiniMax H3)', which contains a detailed prompt and various settings. The prompt describes a scene of a person planting a seedling in a garden. The settings include width (608), height (352), duration (4.0), and a seed (757358688076807). The workflow also includes a 'Resolution Selector' node and a 'Save Video' node. The 'Save Video' node shows the output video player with a 4-second clip.

Resolution Selector

megapixels	Aspect	Output (multiple of 32)
0.2	16:9	608 x 352
0.3	16:9	736 x 416
0.4	16:9	864 x 480
0.5	16:9	992 x 544
0.6	16:9	1120 x 608
0.7	16:9	1248 x 672
0.8	16:9	1376 x 736
0.9	16:9	1504 x 800

Note: Size Settings Reference

Image to Video (MiniMax H3)

Promotional high-end documentary video in a small, maintained community garden on a bright but natural day. The camera is completely static and locked in one fixed position. The ending is a medium close-up of a person's hands planting a seedling in soil, with a garden hose and a small container of finished compost visible nearby. The garden is lush and green, with various plants and flowers. The person is wearing a blue shirt and is focused on the task. The overall tone is peaceful and hopeful, emphasizing the importance of sustainable living and community gardens.

Save Video

video/MiniMax_H3

format: auto

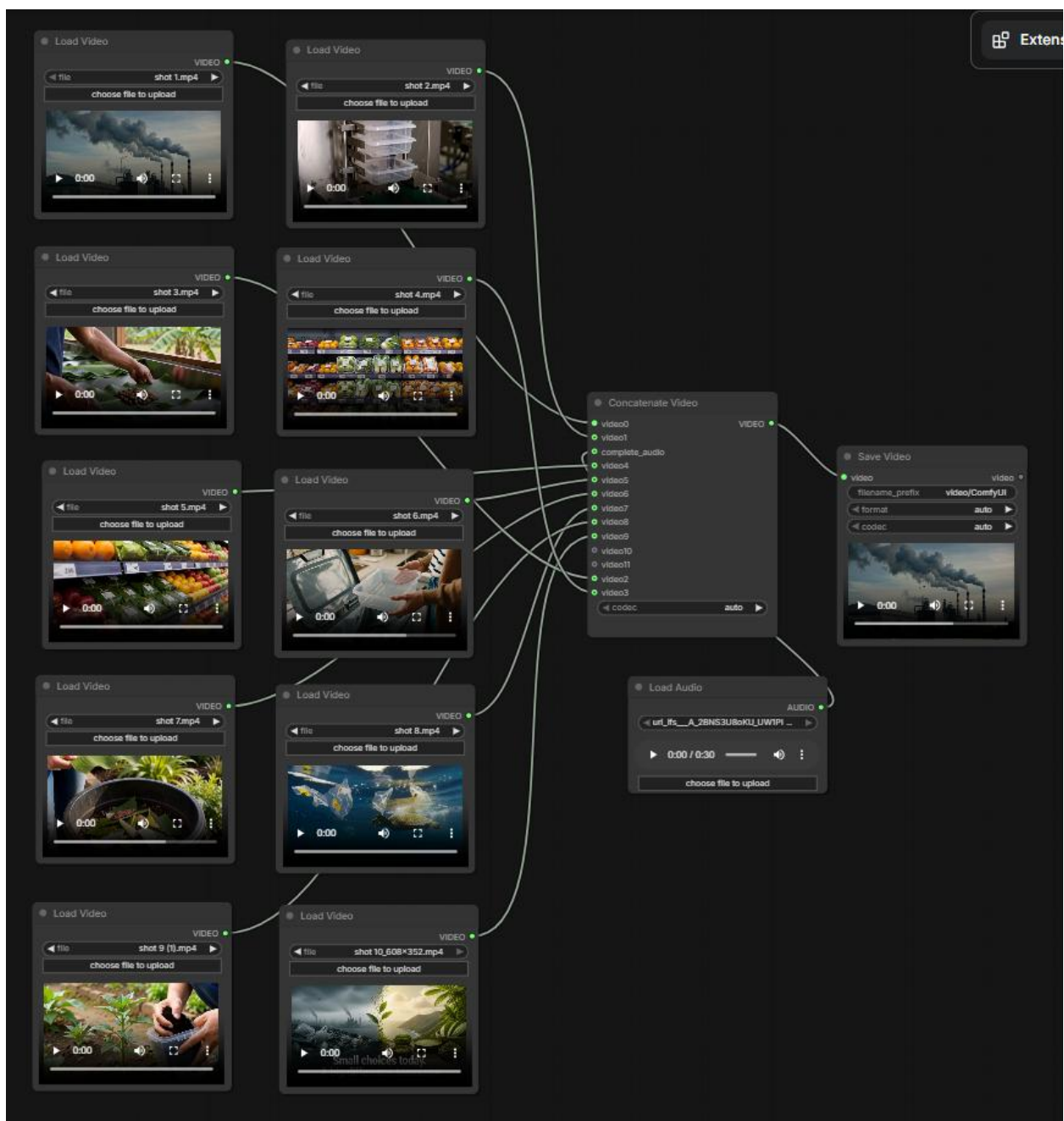
codec: auto

0:00 / 0:04

2. Video Assembly & Audio Muxing Pipeline (Concatenate Workflow)

This workflow connects all ten individually generated ComfyUI video clips into a single cohesive 30-second climate short film, fully synchronized with the master soundtrack.

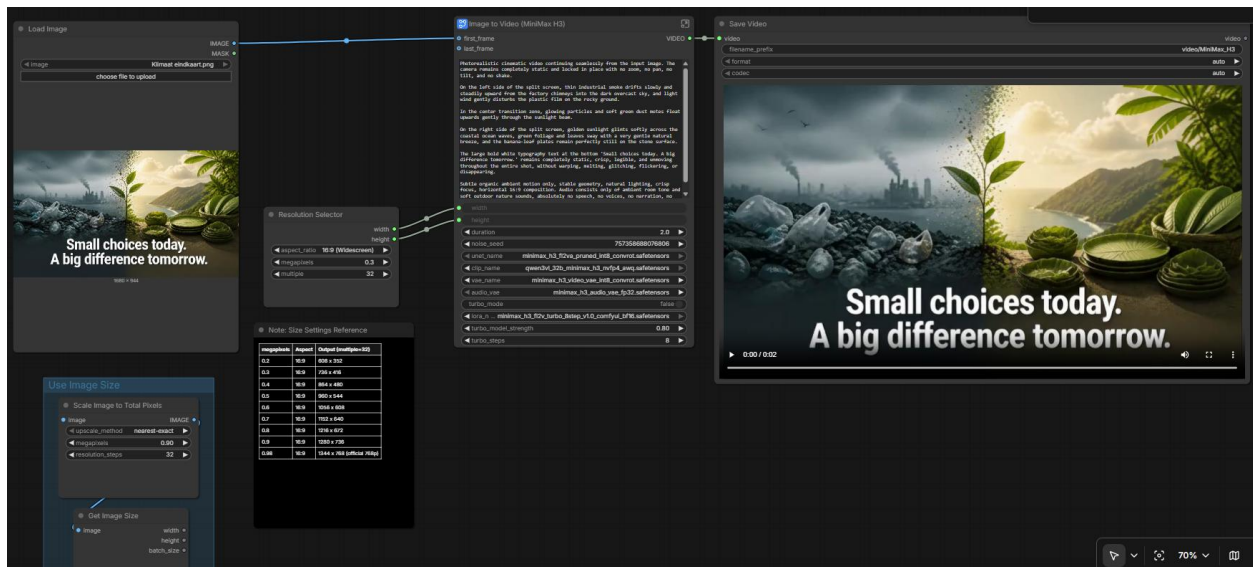
- **Nodes Used:** Load Video (10 instances, sequentially ingesting shot 1.mp4 through shot 10_608x352.mp4 as video buffers); Load Audio (importing the royalty-free background soundtrack and ambient foley audio); Concatenate Video (stitching the 10 video inputs chronologically across video0 through video9 with strict 608×352 dimension and framerate parity); Save Video (exporting the final compiled 30-second film directly from the ComfyUI canvas).
- **Rubric & Assessment Relevance (K1 / Solution Description):** Demonstrates that post-processing, sequence assembly, and audio synchronization were completely integrated inside native ComfyUI graph operations rather than utilizing external non-verifiable video editors.



3. Conditioned End-Card Generation (Image-to-Video Pipeline)

This workflow utilizes Image-to-Video conditioning from a 16:9 concept split-screen graphic (Klimaat eindkaart.png) as a first_frame input to create a smooth, animated conclusion displaying the campaign motto.

- **Nodes Used:** Load Image (ingests the high-contrast source graphic showing industrial pollution versus regenerative nature and the concluding motto 'Small choices today. A big difference tomorrow.');
- Resolution Selector (routes exact pixel dimensions directly to the sampler);
- Image to Video (MiniMax H3) (animates subtle background elements such as chimney smoke and sunlight glints while locking typography sharpness via prompt and denoise guidance);
- Save Video (encodes the 2-second animated end-card for the film outro).
- **Rubric & Assessment Relevance (K3 / Ethical Reasoning & Prototype Fit):** Shows advanced Image-to-Video conditioning in ComfyUI, demonstrating precise control over generative stability to deliver a clear, legible climate message without text degradation.



Shot 1 prompt

Photorealistic cinematic documentary-style video of a fictional industrial complex involved in the production of raw materials for plastic packaging. Exterior establishing shot at early morning. The camera is completely static, positioned at a safe distance across an industrial access road. The wide horizontal frame shows the entire factory complex: large production buildings, metal pipes, processing towers, and several tall stacks rising into a dark overcast sky.

A large amount of visible grey industrial emissions rises continuously from the stacks. The emissions move upward and drift slowly in one consistent direction with the wind. They remain connected to the stacks throughout the shot. The plumes gradually expand and disperse naturally; they do not form impossible shapes, abruptly change direction, flicker, disappear, or cover the entire image. Keep the scene dramatic but physically believable, with no fire, explosion, emergency, or catastrophic event.

In the foreground, a single industrial truck drives slowly along the access road toward the factory. Its wheels remain in contact with the road and rotate naturally. The truck maintains the same shape, color, and size as it moves. Factory buildings, pipes, towers, stacks, fences, and road markings remain fixed and geometrically consistent from beginning to end.

The atmosphere is serious and subdued. Cool grey-blue color grading, diffused morning light, slight atmospheric haze, realistic metal and concrete textures, accurate shadows, natural reflections, and believable depth of field. The image should look like professionally filmed footage of a real industrial site, while remaining a fictional visual representation rather than footage of an identifiable company or location.

The camera remains completely stationary throughout the shot. No zoom, pan, tilt, rotation, handheld shake, perspective change, or scene cuts. Continuous realistic motion only in the drifting emissions and slowly moving truck. No duplicated vehicles, morphing buildings, unstable geometry, flickering objects, or sudden changes in lighting.

Audio consists only of distant industrial ambience: low machinery rumble, ventilation noise, a faint truck engine, and wind. No people in close-up, no voices, speech, whispering, narration, dialogue, music, subtitles, readable signs, company names, or logos.

Shot 2 prompt

Photorealistic high-end industrial documentary video inside a modern, clean automated food packaging factory. The camera is completely static and locked in one fixed position, showing a stainless-steel vertical de-nesting machine and the green conveyor belt directly underneath it. The framing is a medium close-up focused on the exact point where the plastic food trays are dispensed onto the conveyor.

The de-nesting machine contains a vertical stack of identical clear rigid rectangular plastic food trays. Each tray is a shallow transparent food container with a perfectly flat bottom, straight rectangular edges, consistent thickness, and realistic rigid plastic material. The trays must remain physically solid

and maintain exactly the same shape, dimensions, transparency, and orientation throughout the entire video.

The automated mechanism operates continuously and precisely. One single tray at a time is separated from the bottom of the vertical stack by the stainless-steel de-nesting mechanism. The tray is mechanically lowered and released directly above the moving green conveyor belt. The tray falls only a very short realistic distance under gravity, lands flat and naturally on the conveyor belt, and immediately becomes supported by the belt.

After landing, the tray remains completely flat, stable, horizontal, and correctly aligned with the conveyor. The conveyor moves continuously in one single direction at a constant realistic industrial speed, carrying the tray smoothly away from the dispensing mechanism. The tray must move together with the conveyor surface without floating, sliding independently, bouncing, rotating, flipping, tumbling, tilting, bending, deforming, melting, collapsing, or changing shape.

The machine then repeats the same precise cycle: the next tray is separated from the stack, lowered, released, lands flat on the conveyor, and is transported forward. Only one tray is released at a time. Every tray remains clearly separated from the other trays with realistic spacing. Trays never overlap, merge, pass through each other, duplicate, disappear, or suddenly change position.

All mechanical components behave according to realistic industrial engineering and physics. Stainless-steel guides, brackets, rollers, pneumatic actuators, sensors, servo motors, supports, and moving parts remain physically connected and maintain their correct geometry. The mechanical movement is smooth, controlled, repetitive, and consistent from the beginning to the end of the video. No mechanical parts morph, stretch, disappear, or move unnaturally.

The green conveyor belt has realistic rubber texture and subtle surface detail. The stainless-steel machinery has realistic polished-metal reflections, accurate highlights, natural shadows, and physically plausible reflections. The clear plastic trays have realistic transparent plastic reflections and refraction while remaining clearly visible against the conveyor belt.

The factory environment is clean, modern, organized, and realistic, with professional industrial lighting and natural ambient illumination. The scene should look like genuine footage recorded inside a real automated packaging facility rather than computer-generated imagery. Extremely detailed materials, realistic textures, accurate proportions, realistic depth of field, crisp focus, high dynamic range, natural motion blur, physically accurate lighting, and 8K-quality visual detail.

The camera remains completely stationary throughout the entire shot with no camera movement, no handheld shake, no zoom, no pan, no tilt, no rotation, no perspective change, and no scene cuts. The de-nesting mechanism and tray transfer point remain clearly visible and consistently framed for the entire duration.

The video contains continuous realistic industrial motion with stable temporal consistency. Objects retain their identity, shape, position, material, and physical properties from frame to frame. There are

no sudden changes, visual glitches, flickering objects, unstable geometry, duplicated objects, disappearing objects, or temporal inconsistencies.

Audio consists only of authentic factory ambience: a low continuous conveyor motor hum, subtle electric servo motor whirring, quiet pneumatic air hiss when each tray is released, soft mechanical clicks from the de-nesting mechanism, and a subtle conveyor belt hum. No people, no workers, no voices, no talking, no speech, no whispering, no narration, no dialogue, no words, no subtitles, no music, and no dramatic sound effects.

The final result must look like authentic professional industrial documentary footage of a real automated food packaging line, with physically accurate machinery, realistic gravity and collisions, stable rigid plastic trays, smooth conveyor movement, consistent object geometry, and precise repeatable automation.

Shot 3 prompt

Photorealistic high-end documentary video inside a small, clean food-packaging workshop near a banana plantation. The camera is completely static and locked in one fixed position. The framing is a medium close-up of a stainless-steel worktable, focused on a worker's hands carefully folding one large banana leaf around a suitable dry food product. The worker's face and body are outside the frame.

One clean, intact banana leaf lies flat on the worktable at the beginning of the shot. It has realistic green color, visible natural veins, a slightly flexible surface, and consistent size and shape. The dry food product rests securely in the center of the leaf. The worker uses both hands to fold the left and right sides inward, then folds the remaining edge over the product to form a simple, practical wrap.

The hands move slowly and deliberately. The leaf bends naturally along the folds without tearing, stretching, melting, or changing into another material. The food product remains in the center and does not disappear, move through the leaf, or change shape. The wrap stays on the worktable throughout the shot. Show only this one wrapping action; do not suddenly switch to harvesting, shipping, supermarket shelves, or composting.

The workshop is clean and believable, with stainless-steel surfaces, a few neatly stacked banana leaves, and soft daylight entering from a window. Banana plants are visible outside in the background but remain stationary except for subtle natural movement in the breeze. The scene represents one possible packaging method, not a claim that banana leaves work for every food product.

Realistic materials, accurate hand anatomy, physically believable folds, natural shadows, stable object geometry, consistent lighting, crisp focus on the hands and leaf, and subtle depth of field. The footage must look like a real documentary recording, not an animation or a commercial fantasy.

The camera does not move, zoom, pan, tilt, rotate, or cut. The leaf, product, hands, table, and background retain their identity and position from frame to frame. No extra hands appear. No leaves duplicate, vanish, float, or fold by themselves.

Audio consists only of quiet workshop ambience, soft leaf rustling, subtle hand contact with the worktable, and distant outdoor birds. No speech, narration, dialogue, music, subtitles, readable text, branding, or dramatic sound effects.

Shot 4 prompt

Photorealistic high-end documentary video inside a modern European supermarket. The camera is completely static and locked in one fixed position, facing a single produce shelf at eye level. The framing is a medium-wide shot showing several rows of ordinary fruits and vegetables. Many individual items are unnecessarily wrapped in separate clear plastic trays or transparent plastic film.

The products are arranged neatly and realistically. Clear rigid plastic trays maintain their shape and transparency. Transparent wrapping remains attached to the correct products and does not float, ripple unnaturally, merge with other packaging, or change material. The fruits and vegetables have believable sizes, textures, and colors. The shelf looks like a real supermarket display, not an exaggerated wall of plastic.

During the shot, one shopper's hand enters the frame from the right, picks up one individually packaged product from the middle shelf, and removes it from the frame. The hand has natural anatomy and makes believable contact with the package. Only that one product moves. All other products and packages remain stable in their original positions. The shelf does not suddenly refill itself.

The lighting is neutral, slightly cool supermarket lighting. The image has a restrained, observational mood. The camera angle and framing should be easy to match with a second version of the same shelf showing less packaging. Do not show a dramatic transformation within this shot.

Realistic supermarket materials, physically accurate transparent plastic reflections, natural shadows, consistent shelf geometry, crisp product detail, and stable temporal consistency. No warped packaging, duplicated products, changing labels, distorted hands, or flickering objects.

The camera remains completely stationary with no zoom, pan, tilt, rotation, handheld shake, perspective change, or cuts. Audio consists only of subtle supermarket ambience, distant trolley wheels, soft ventilation, and the quiet sound of the package being lifted. No voices, speech, narration, dialogue, music, subtitles, readable labels, prices, logos, or branding.

Shot 5 prompt

Photorealistic high-end documentary video inside a modern European supermarket. The camera is completely static and locked in one fixed position, facing a produce shelf at eye level. Match the medium-wide framing, camera height, shelf layout, and viewing angle of the previous supermarket shot as closely as possible.

The shelf displays comparable fruits and vegetables loose where this is practical. A small selection of suitable dry food products is wrapped in simple, neatly folded banana-leaf packaging. Do not put banana

leaves around every product. The display must look like a believable supermarket shelf rather than a fantasy store. Leaves have natural green color, visible veins, realistic folds, and slightly varied surfaces.

During the shot, one shopper's hand enters from the right, carefully picks up one banana-leaf-wrapped product from the middle shelf, and removes it from the frame. The hand makes natural contact with the wrap. The folded leaf remains securely around the product and does not unfold, change size, become plastic, or reveal a different object. Only the selected package moves. All other products remain in place.

Warm but realistic supermarket lighting creates a more welcoming atmosphere than the previous shot. The difference between the two shelves should come from the visible packaging choices, not from magical greenery growing inside the supermarket.

Accurate proportions, natural hand anatomy, physically believable leaf texture, stable shelf geometry, realistic depth of field, consistent objects and lighting from frame to frame. No duplicated hands, floating packages, morphing leaves, disappearing products, unreadable generated labels, or sudden changes in shelf layout.

The camera remains completely stationary with no zoom, pan, tilt, rotation, handheld shake, perspective change, or cuts. Audio consists only of quiet supermarket ambience, distant trolley wheels, soft ventilation, and gentle leaf rustling when the package is picked up. No voices, speech, narration, dialogue, music, subtitles, readable prices, logos, or branding.

Shot 6 prompt

Photorealistic high-end documentary video inside an ordinary modern kitchen. The camera is completely static and locked in one fixed position at approximately waist height. The framing is a medium close-up of a person's hands, an empty clear rigid plastic food tray, and an open household residual-waste bin. The person's face is never visible.

At the beginning of the shot, the person holds one empty plastic tray above the bin. The tray has a flat bottom, straight rectangular edges, consistent transparency, and realistic rigid plastic thickness. The person releases the tray once. It falls a short distance under gravity, lands inside the bin, and comes to rest among a modest amount of ordinary household waste.

The tray stays physically solid during the entire action. It must not bend, melt, duplicate, disappear, float, pass through the side of the bin, bounce back out, or change into another object. The person's hands move naturally away after releasing it. The bin lid then closes gently. Show only this one disposal action.

The kitchen is realistic, clean, and lived-in. Use neutral daylight with a slightly subdued mood. Do not show an enormous landfill or an incinerator suddenly appearing behind the kitchen. The shot should feel like a normal daily action whose consequences are easy to overlook.

Accurate gravity and contact, natural hand anatomy, stable plastic geometry, realistic reflections on the clear tray, consistent bin position, natural shadows, and stable temporal consistency. The camera remains stationary with no zoom, pan, tilt, rotation, shake, perspective change, or scene cuts.

Audio consists only of quiet kitchen room tone, the light contact of the tray landing in the bin, and the soft sound of the lid closing. No voices, speech, whispering, narration, dialogue, music, subtitles, readable text, logos, or dramatic sound effects.

Shot 7 prompt

Photorealistic high-end documentary video beside a maintained home compost container in a private garden. The camera is completely static and locked in one fixed position at approximately waist height. Match the camera distance and framing of the previous plastic-disposal shot. Show a person's hands, one used uncoated banana-leaf wrap, and the open compost container. The person's face is never visible.

At the beginning of the shot, the person holds one used banana-leaf wrap above the compost container. The leaf is visibly the same type of simple folded wrap shown earlier. It remains recognizable as a natural leaf, with visible veins, a flexible surface, and realistic folds. The person places it gently among other appropriate plant scraps inside the container, then withdraws their hands.

The banana leaf does not instantly decompose. It does not turn into soil during this shot, grow into a plant, disappear, float away, or change into another material. The compost container is clearly a managed place for organic material, not a random area of forest or public nature. Nothing is thrown on the ground.

Warm natural daylight illuminates the hands, leaf, and container. Plants in the background move only slightly in a breeze. The mood is calm and practical rather than magical. Show one clear disposal action from beginning to end.

Realistic hand anatomy, believable leaf flexibility, physically accurate contact with the compost contents, natural soil and plant textures, stable object positions, consistent shadows, and temporal continuity. No extra hands, duplicated leaves, sudden changes in the compost container, or objects passing through each other.

The camera remains completely stationary with no zoom, pan, tilt, rotation, shake, perspective change, or cuts. Audio consists only of soft outdoor ambience, gentle leaf rustling, and a light breeze. No voices, speech, narration, dialogue, music, subtitles, readable text, logos, or dramatic sound effects.

Shot 8 prompt

Photorealistic high-end industrial documentary video outside a real-looking municipal waste-processing facility on an overcast day. The camera is completely static and locked in one fixed position. The framing

is a medium-wide shot showing the entrance road in the foreground and the facility in the background. One waste-collection truck approaches the entrance slowly and continues forward at a realistic speed.

The facility has plausible industrial architecture, loading doors, service roads, safety barriers, and distant ventilation structures. Everything remains correctly connected and consistent from frame to frame. Do not show a factory engulfed in black smoke, open flames, explosions, or an apocalyptic sky. The purpose of the shot is to show that discarded packaging continues into an industrial waste system, not to portray a specific real facility or claim that all waste is processed identically.

The truck stays the same size and shape throughout the shot. Its wheels rotate naturally and remain in contact with the road. It travels in one direction without teleporting, duplicating, changing model, or passing through barriers. No other vehicles suddenly appear or disappear. The building, signs, doors, and background remain stable.

Cool natural daylight, subdued color grading, believable concrete and metal textures, physically plausible reflections, accurate shadows, crisp focus, and a restrained documentary mood. The footage should resemble a real observational shot filmed from a safe distance.

The camera remains completely stationary with no zoom, pan, tilt, rotation, handheld shake, perspective change, or cuts. Audio consists only of distant industrial ambience, a low ventilation hum, and the approaching truck's engine and tires. No voices, speech, narration, dialogue, music, subtitles, readable signs, logos, or dramatic sound effects.

Shot 9 prompt

Photorealistic high-end documentary video in a small, maintained community garden on a bright but natural day. The camera is completely static and locked in one fixed position. The framing is a medium close-up of a young garden plant growing in soil, with a gardener's hands and a small container of finished compost visible beside it. The gardener's face is outside the frame.

The gardener takes one small handful of already finished, dark, crumbly compost from the container and gently spreads it on the soil around the base of the plant. The compost falls naturally in small pieces under gravity and settles on the ground. The hands then move away, leaving the plant clearly visible. Show only this single gardening action.

The plant does not grow instantly, bloom suddenly, change species, or move unnaturally. The compost does not magically appear from a banana leaf during this shot. This scene represents a later stage of a managed composting process, not an immediate transformation of one used package into soil.

The garden contains believable planting beds, natural soil, a few healthy plants, and soft background greenery. Warm daylight creates a hopeful mood without making the scene look like a fantasy. Leaves move subtly in the breeze. The soil, compost, hands, and plant maintain their physical form and position from frame to frame.

Realistic hand anatomy, natural compost texture, accurate gravity and contact with the soil, consistent plant geometry, soft natural shadows, detailed materials, crisp focus, gentle depth of field, and stable temporal consistency. No duplicated hands, floating soil, flickering leaves, disappearing objects, or sudden environmental changes.

The camera remains completely stationary with no zoom, pan, tilt, rotation, shake, perspective change, or cuts. Audio consists only of a soft breeze, distant birds, light movement of leaves, and subtle contact between compost and soil. No voices, speech, narration, dialogue, music, subtitles, readable text, logos, or dramatic sound effects.

Shot 10 prompt

Photorealistic cinematic video continuing seamlessly from the input image. The camera remains completely static and locked in place with no zoom, no pan, no tilt, and no shake.

On the left side of the split screen, thin industrial smoke drifts slowly and steadily upward from the factory chimneys into the dark overcast sky, and light wind gently disturbs the plastic film on the rocky ground.

In the center transition zone, glowing particles and soft green dust motes float upwards gently through the sunlight beam.

On the right side of the split screen, golden sunlight glints softly across the coastal ocean waves, green foliage and leaves sway with a very gentle natural breeze, and the banana-leaf plates remain perfectly still on the stone surface.

The large bold white typography text at the bottom 'Small choices today. A big difference tomorrow.' remains completely static, crisp, legible, and unmoving throughout the entire shot, without warping, melting, glitching, flickering, or disappearing.

Subtle organic ambient motion only, stable geometry, natural lighting, crisp focus, horizontal 16:9 composition. Audio consists only of ambient room tone and soft outdoor nature sounds, absolutely no speech, no voices, no narration, no extra sound effects.

